5 Grammar for Regular expression: (1)

$0^* 1 (0+1)^*$

Grammar:

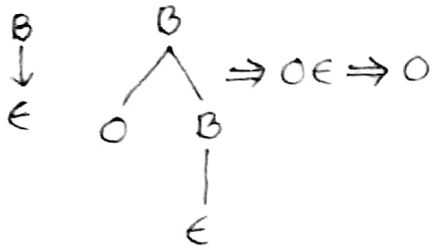
$S \longrightarrow A\ 1\ B$

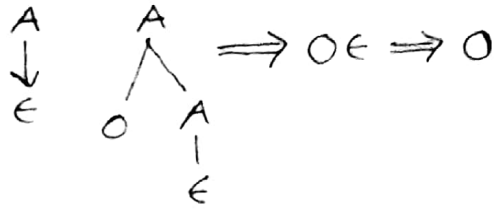
$A \longrightarrow \epsilon \mid 0A$

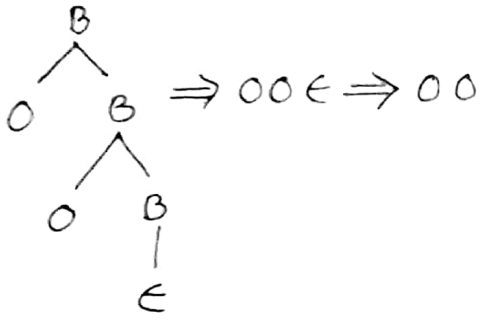
$B \longrightarrow \epsilon \mid 0B \mid 1B$

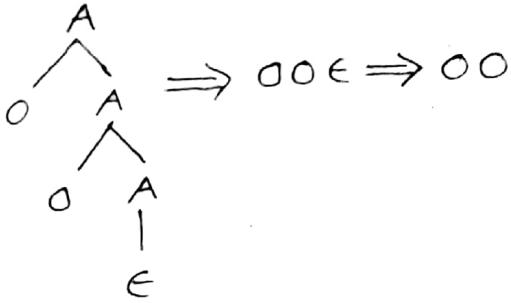
$A = 0^* = \{\epsilon, 0, 00, 000, 0000, \cdots\}$

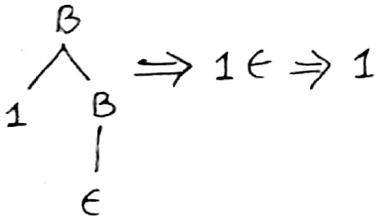
$B = (0+1)^* = \{\epsilon, 0, 1, 01, 10, 00, 11, 000, 111, \cdots\}$

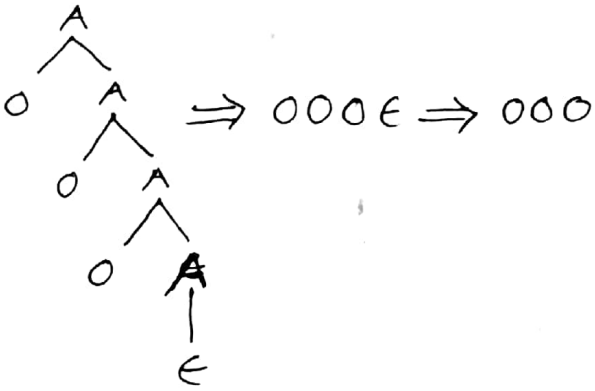


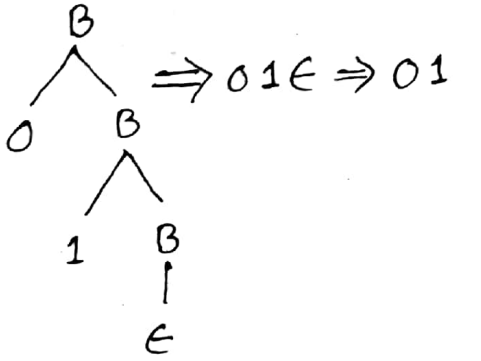


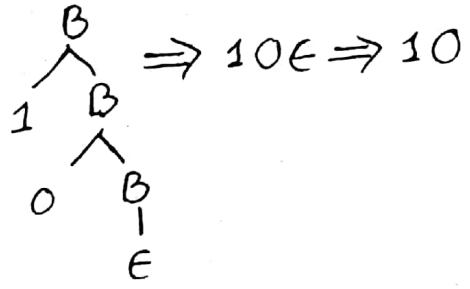












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$DD^* = D^+$ (Regular expression for positive integer numbers)

$DD^* = D\{\epsilon, D, DD, DDD, \ldots\}$

$= \{D\epsilon, DD, DDD, DDDD, \ldots\}$

$= \{D, DD, DDD, DDDD, \ldots\}$

$= D^+$

$D = 0 | 1 | 2 | \cdots | 9$

For example,

$D = 8$

$DD = 19$

$DDD = 100$

$DDDD = 1009$

Regular expression for unsigned decimal numbers:

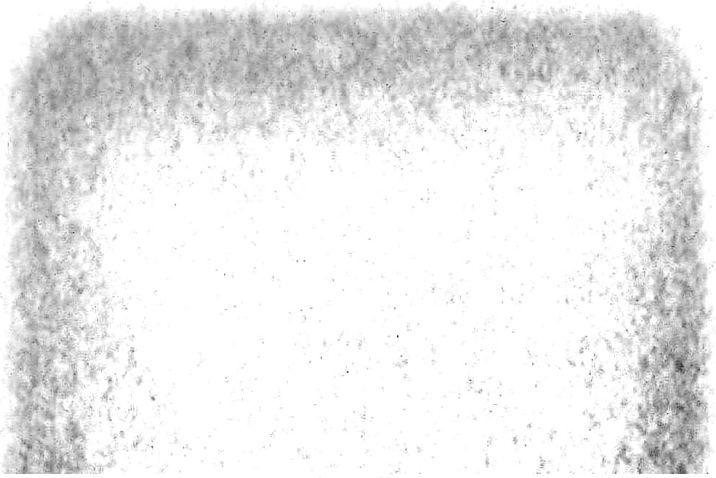
10, 20, 10.5, 20.0, 30, 30.65

$RE = D^+(\epsilon | .D^+)$

$= D^+\epsilon | D^+.D^+$ For integer only D^+

$= D^+ | D^+.D^+$ For fractional numbers $D^+.D^+$

Here, $D^+ = \{D, DD, DDD, DDDD, \ldots\}$



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